

Chapter 2

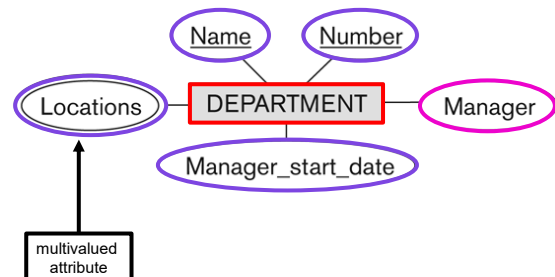
Data Modeling Using the Entity-Relationship (ER) Diagram

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The COMPANY Database

We need to design a database schema for the COMPANY database based on the following requirements:

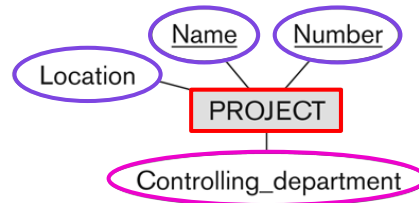
- The company is organized into **DEPARTMENTS**.
- Each department has a unique **name** and unique **number**.
- Each department has an employee who **manages** the department.
- We record the **start date** of the department manager.
- A department may have several **locations**.



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The company performs multiple **PROJECT**s

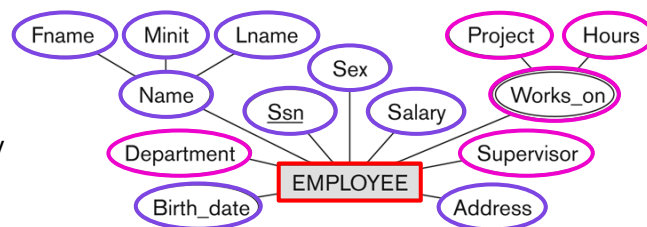
- Each project is **controlled** by some **department**.
- Each project has a unique **name** and unique **number**.
- Each project is located at a single **location**.



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The company employs multiple **EMPLOYEE**s. Each employee has:

- A **name**, consisting of first name, middle initial and last name.
- A **social security number**, **address**, **salary**, **sex**, and **birthdate**.
- **works for** one **department** but may **work on** several **projects**.
- works a number of **hours** per week on each project.
- has a **direct supervisor**.



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ER Model Concepts

Entities

Entities are specific things or objects in the mini-world that are represented in the database.

the DEPARTMENT

the PROJECT

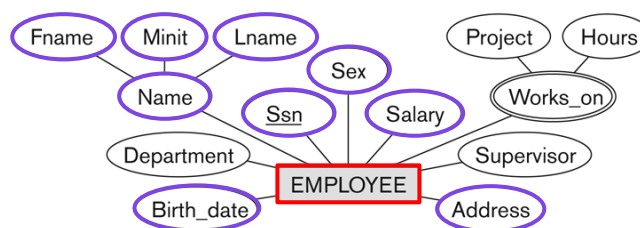
the EMPLOYEE

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ER Model Concepts

Attributes are properties of an entity.

For example, an EMPLOYEE entity may have the attributes Name, Ssn, Address, Sex, Birthdate, Salary



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ER Model Concepts

An entity instance will have a value for each of its attributes.

E.g. a specific employee may have

- **Name** = 'John R. Smith',
- **SSN** = 123456789,
- **Address** = '731 Fondren Rd., Houston, TX',
- **Sex** = M,
- **Birthdate** = '01-09-1975'

Each attribute has a **value set** (or data type) associated with it –
e.g. integer, string, date, enumerated type, ...

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Types of Attributes

Simple

Each **entity** has a single atomic value for the **attribute**. For example,
Ssn or **Sex**.

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Types of Attributes

Composite

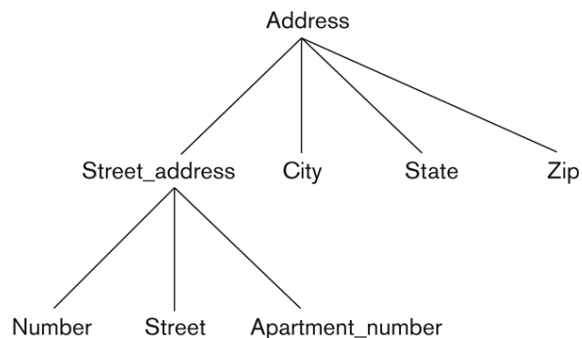
The **attribute** may be composed of several components. For example:

- **Address** (Apt#, House#, Street, City, State, ZipCode, Country)
- **Name** (FirstName, MiddleName, LastName)
- Composition may form a hierarchy where some components are themselves composite.

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Composite Attribute

A hierarchy of composite **attributes**



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Types of Attributes

Multi-valued

An **entity** may have multiple values for that **attribute**. For example, **Color** of a **CAR** or **PreviousDegrees** of a **STUDENT**.

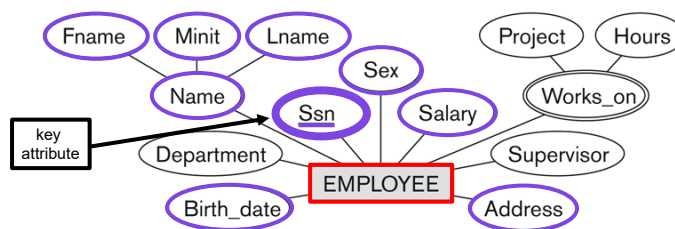
- Denoted as {Color} or {PreviousDegrees}.

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Key Attributes

An **attribute** of an entity for which each **entity** must have a **unique value** is called a **key attribute** of the entity.

E.g. **Ssn** of **EMPLOYEE**.



The **key** is underlined

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Key Attributes

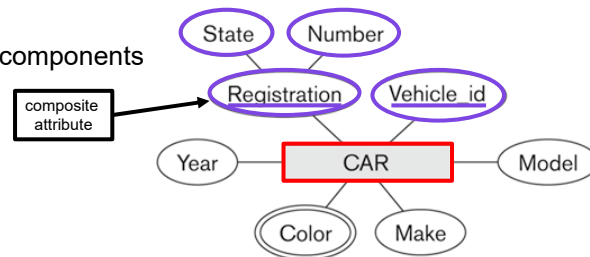
A key **attribute** may be composite.

E.g. **Registration** is a key of **CAR** with components
(**Number**, **State**).

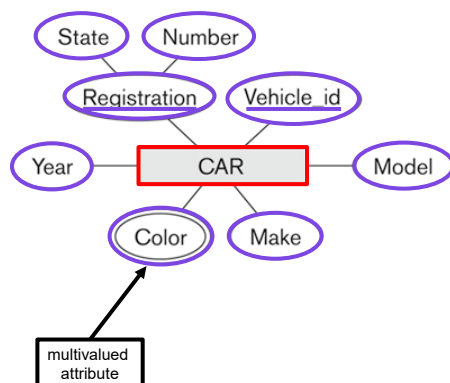
An **entity** may have more than one key.

CAR may have two keys:

- **Registration**
- **VehicleId** (AKA VIN)



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CAR
Registration (Number, State), Vehicle_id, Make, Model, Year, {Color}

CAR₁
((ABC 123, TEXAS), TK629, Ford Mustang, convertible, 2004 {red, black})

CAR₂
((ABC 123, NEW YORK), WP9872, Nissan Maxima, 4-door, 2005, {blue})

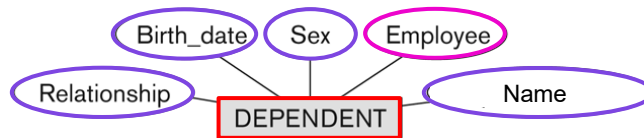
CAR₃
((VSY 720, TEXAS), TD729, Chrysler LeBaron, 4-door, 2002, {white, blue})

⋮

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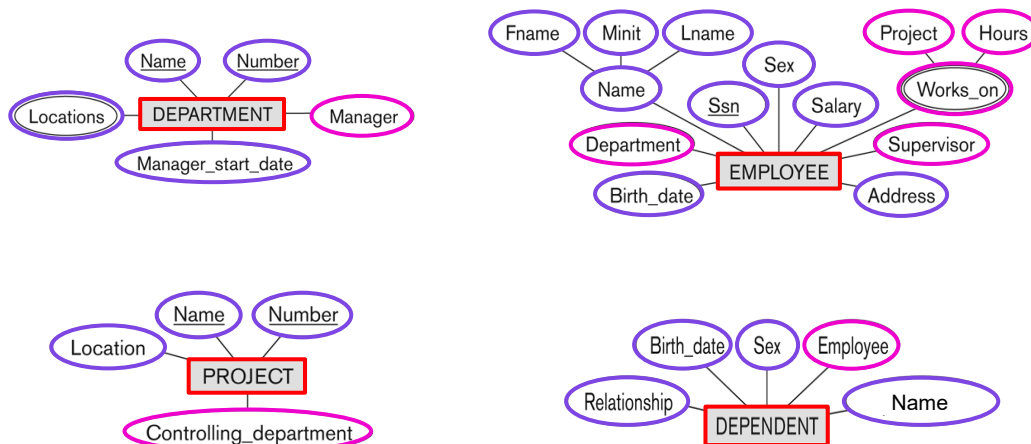
More Entities

We also want to keep track of the **DEPENDENTS** of each **EMPLOYEE** for insurance purposes. We keep each **DEPENDENT's** name, sex, birth date, and relationship to the **EMPLOYEE**.



What is the key attribute of **DEPENDENT** ?

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Relationships

An ER model has three main components:

- **Entities**
- **Attributes** (simple, composite, multivalued)
- **Relationships**

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Relationships and Relationship Types

A **relationship** relates two or more distinct **entities**

For example,

EMPLOYEE John Smith **works for** the Research **DEPARTMENT**

EMPLOYEE John Smith **works on** the Engine **PROJECT**

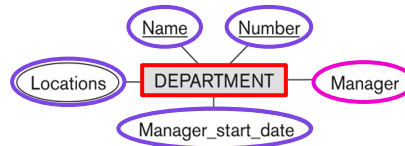
EMPLOYEE Franklin Wong **manages** the Research **DEPARTMENT**.



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The attribute **Manager** of **DEPARTMENT** refers to an **EMPLOYEE** who manages the **DEPARTMENT**.



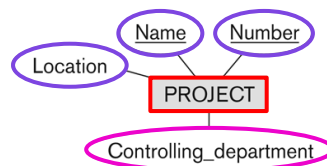
MANAGES relates **EMPLOYEE** and **DEPARTMENT**



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The attribute **Controlling_department** of **PROJECT** refers to the **DEPARTMENT** that controls the **PROJECT**.



CONTROLS relates **PROJECT** and **DEPARTMENT**



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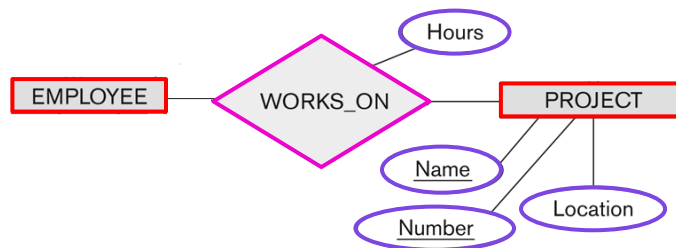
WORKS_FOR relates DEPARTMENT and EMPLOYEE



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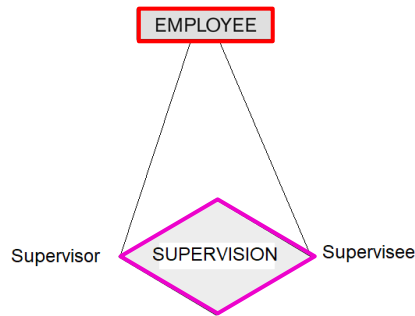
WORKS_ON relates EMPLOYEE and PROJECT



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SUPERVISION relates **EMPLOYEE** (in the supervisor role)
and **EMPLOYEE** (in the supervisee role).



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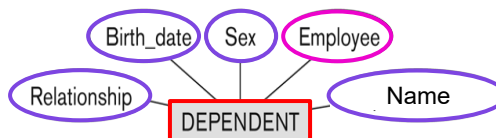
Weak Entities

An entity that does not have a key attribute and that is identification-dependent on another entity.

A weak entity must participate in an *identifying relationship* with an owner or *identifying entity*

Entities are identified by the combination of:

- A partial key of the weak entity
- The particular entity they are related to in the identifying relationship

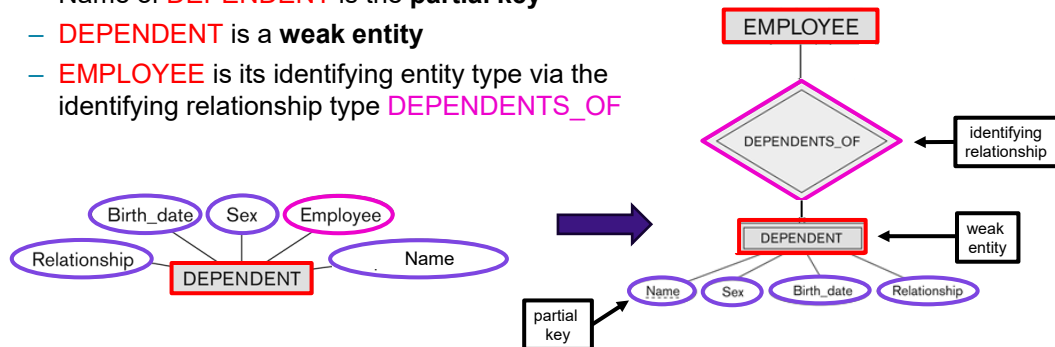


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Weak Entities

Example:

- The **DEPENDENT** entity is identified by the dependent's name, **and** the specific **EMPLOYEE** with whom the **DEPENDENT** is related
- Name of **DEPENDENT** is the **partial key**
- **DEPENDENT** is a **weak entity**
- **EMPLOYEE** is its identifying entity type via the identifying relationship type **DEPENDENTS_OF**



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By examining the requirements, six **binary** relationships are identified:

WORKS_FOR (between **EMPLOYEE** and **DEPARTMENT**)

MANAGES (between **EMPLOYEE** and **DEPARTMENT**)

CONTROLS (between **DEPARTMENT** and **PROJECT**)

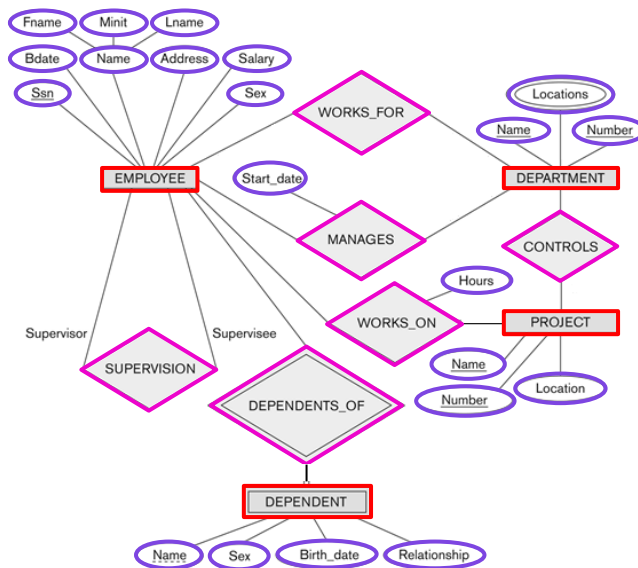
WORKS_ON (between **EMPLOYEE** and **PROJECT**)

SUPERVISION (between **EMPLOYEE** (as subordinate) and **EMPLOYEE** (as supervisor))

DEPENDENTS_OF (between **EMPLOYEE** and **DEPENDENT**)

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An starter ER diagram for the COMPANY database



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One-to-One (1:1) Relationship

An **EMPLOYEE** can manage at most one **DEPARTMENT**.

A **DEPARTMENT** can have at most one managing **EMPLOYEE**.

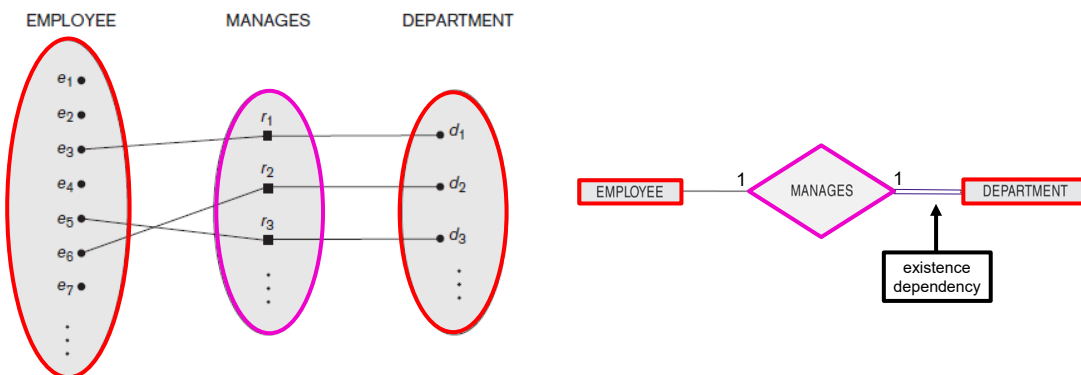


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Constraints on Relationships

Existence Dependency or Total Participation:

A **DEPARTMENT** must have a managing **EMPLOYEE**



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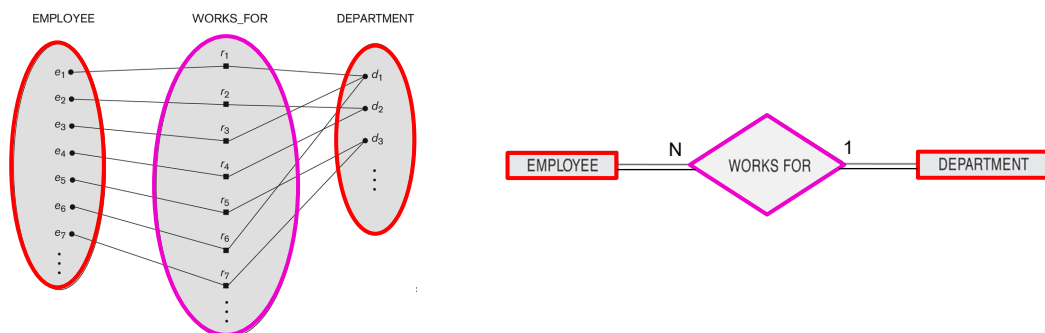
Many-to-One (N:1) Relationship

A **DEPARTMENT** can employ any number of **EMPLOYEE**s (N).

An **EMPLOYEE** can work for at most one **DEPARTMENT** (1).

(A **DEPARTMENT** MUST have some **EMPLOYEE**).

(An **EMPLOYEE** MUST work for some **DEPARTMENT**).



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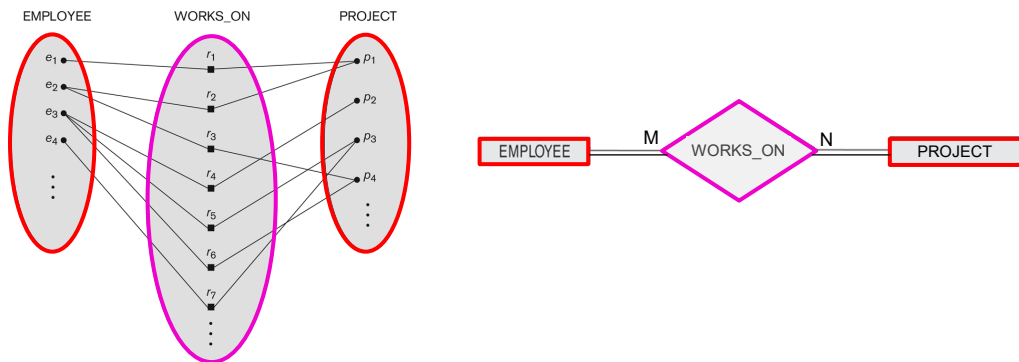
Many-To-Many (M:N) Relationship

A **PROJECT** can employ any number of **EMPLOYEE**s (M).

An **EMPLOYEE** can work on multiple **PROJECT**s (N).

(A **PROJECT** MUST have some **EMPLOYEE**s working on it).

(An **EMPLOYEE** MUST work on some **PROJECT**).

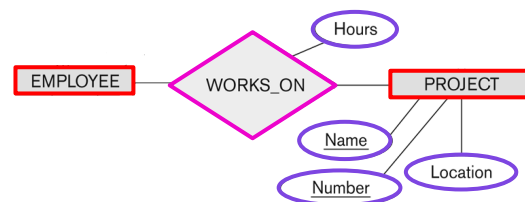


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Attributes of Relationships

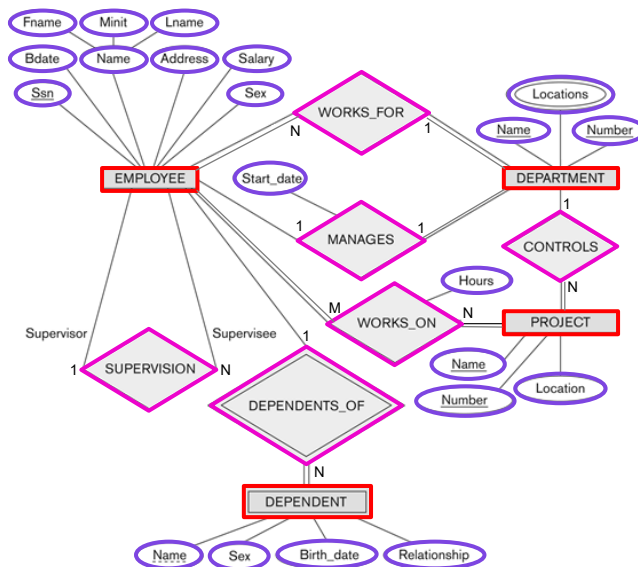
A **relationship** can have **attributes**:

- For example, **Hours** of **WORKS_ON** describes the number of hours per week that an **EMPLOYEE** works on a **PROJECT**.
 - A value of **Hours** depends on a particular (**employee**, **project**) combination
- Most **relationship attributes** are used with M:N **relationships**
 - In 1:N **relationships**, they can be transferred to the **entity** on the N-side of the **relationship**



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An ER diagram for the COMPANY database



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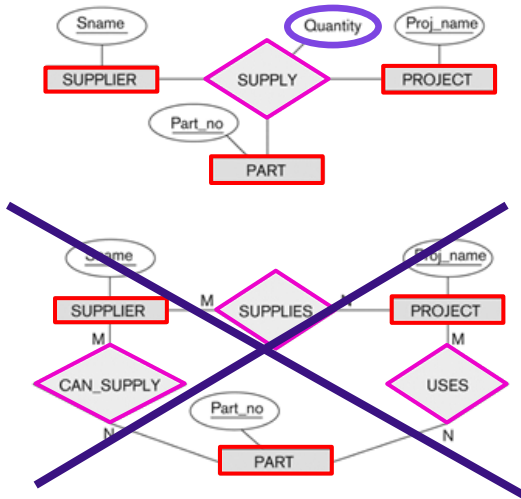
Relationships of Higher Degree

- Relationships of degree 2 are called **binary**
- Relationships of degree 3 are called **ternary**
- Relationships of degree n are called **n-ary**
- An n-ary relationship is **not** equivalent to n binary relationships
- Constraints are harder to specify for higher-degree relationships ($n > 2$) than for binary relationships

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Example of a Ternary Relationship

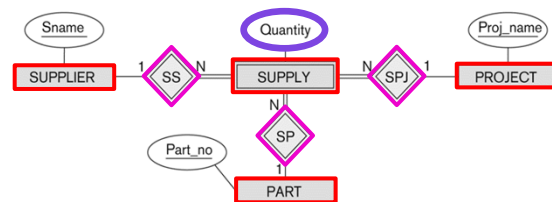
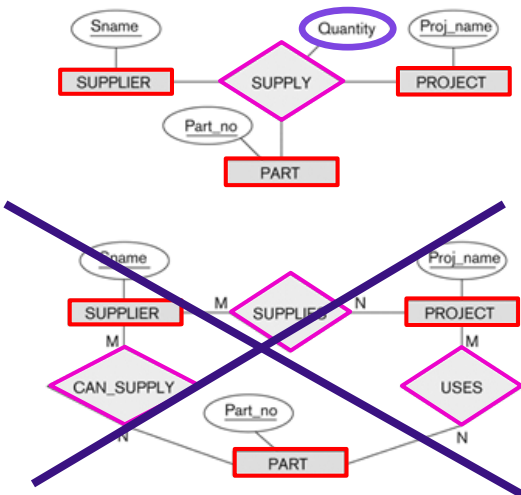
The **SUPPLY** relationship



Three binary relationships
not equivalent to **SUPPLY**

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Simulating a Ternary Relationship



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